

❖ PLANRS ACADEMY — DETAILED LESSON GUIDE ❖

Prime Numbers

Sieve of Eratosthenes

Prime Detectives — Session 1

Subject	Mathematics — Number Theory
Topic	Prime Numbers & Sieve of Eratosthenes
Session	Session 1 of 1 (Stand-alone)
Duration	30 Minutes
Depth	Beginner — Enrichment
Characters	Aarav & Priya (student guides)
Board Alignment	CBSE / ICSE / State Boards (India)
Language	English (child-friendly, bilingual hooks)

LEARNING OBJECTIVES

LO 1	Define a prime number: "exactly 2 factors — 1 and itself."
LO 2	Identify prime numbers from 1 to 30 using the Sieve method.
LO 3	Apply the 5-step sieve process to cross out composites and circle primes.
LO 4	Distinguish between prime and composite numbers using Indian daily-life examples.
LO 5	State that the number 1 is neither prime nor composite (stands alone).
LO 6	Recall Eratosthenes and explain the sieve analogy using atta-sieving.
LO 7	Recognize that 2 is the ONLY even prime number.
LO 8	Express curiosity about prime number patterns (open-ended wonder).

SECTION 1 — THE MYSTERY NUMBER HOOK (0–3 min)

Aarav's Laddoo Problem:

Aarav has **7 laddoos**. Can he share them equally with 2 friends? No! With 3 friends? No! With 6 friends? No! The mysterious number 7 refuses to share equally — except all 7 to 1 person, or 1 each to 7 people. This mysterious number is called a **PRIME NUMBER!** Become a Prime Detective today!

Teacher Script (30–60 sec):

"Who can tell me — if I have 7 laddoos and want to share them equally, how many friends can I give them to?" Accept all answers, try 2, 3, 4, 5, 6 on the board. "Nothing works! 7 is a special, stubborn number — we call it PRIME!"

SECTION 2 — WHAT IS A PRIME NUMBER? (3–8 min)

Core Definition — say aloud AND write on board:

"A prime number has EXACTLY 2 factors: 1 and itself."

Four Properties of Prime Numbers:

Property 1	A prime number can be divided EXACTLY by only two numbers: 1 and itself.
Property 2	Every prime number is greater than 1.
Property 3	The number 2 is the ONLY even prime number.
Property 4	All prime numbers after 2 are ODD numbers.

Indian Context Examples:

■ Rotis 5 rotis on a plate	$5 = 1 \times 5$ or 5×1 only	PRIME ✓
■ Laddoos 7 laddoos in a daba	$7 = 1 \times 7$ or 7×1 only	PRIME ✓
■ Maala 5 beads on a maala	$5 = 1 \times 5$ or 5×1 only	PRIME ✓
■ Mithai 6 mithai in a box	$6 = 1 \times 6, 2 \times 3, 3 \times 2, 6 \times 1$	COMPOSITE ✗

SECTION 3 — THE SPECIAL NUMBER 1 (8–10 min)

The Number 1

Number **1** is **NEITHER prime NOR composite**. It stands alone — unique and different from all other numbers. It has only ONE factor (itself), while prime numbers must have EXACTLY TWO factors.

Memory hook: "Ek toh ek hi hota hai!" — One is always just one! Just like the letter A in the alphabet — unique.

Number	Classification	Working
Number 1	Neither prime nor composite	<i>Stands alone, unique</i>
Number 2	PRIME — first and only even prime	$2 = 1 \times 2$ only
Number 4	COMPOSITE — has 3 factors: 1, 2, 4	$4 = 1 \times 4$ or 2×2
Number 7	PRIME — 2 factors only	$7 = 1 \times 7$ only

SECTION 4 — MEET ERATOSTHENES (10–13 min)

Who was Eratosthenes?

An ancient Greek mathematician from **over 2,200 years ago** invented a brilliant method to find ALL prime numbers at once. He called it a SIEVE — and it still works perfectly today!

Born:

Lived:

Achievement:

Also known for:

■ Date	c. 276–194 BC, ancient Greece
■ Invention	The Sieve method to find ALL prime numbers
■ Indian Link	Just like Daadi sieves atta in the kitchen — primes pass through, composites are removed!
■ Indian Parallel	Aryabhata & Brahmagupta also studied numbers in India — mathematics is a shared human story!

The Atta-Sieve Analogy:

Fine Atta Flour (Passes Through)	=	Prime Numbers (Stay in the list)
Bran & Husks (Caught in sieve)	=	Composite Numbers (Crossed out)

SECTION 5 — HOW THE SIEVE WORKS (13–20 min)

We use the Sieve of Eratosthenes on numbers 1–30. Follow each step carefully. The numbers that survive without being crossed out are ALL prime!

1

Cross out the number 1

Write numbers 1–30 in a grid. Cross out 1 immediately. (Neither prime nor composite)

2

Circle 2 → Cross out all multiples of 2

Circle 2 (first prime!). Cross out: 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30

3

Circle 3 → Cross out remaining multiples of 3

Circle 3 (prime!). Cross out: 9, 15, 21, 27 (6, 12, 18, 24, 30 already done)

4

Circle 5 → Cross out remaining multiples of 5

Circle 5 (prime!). Cross out: 25 (10, 15, 20, 25, 30 mostly done already)

5

Circle all remaining uncrossed numbers

Every remaining number is PRIME! Circle: 7, 11, 13, 17, 19, 23, 29

Primes from 1 to 30 (Visual Grid):

1 (special)	2 ★ prime	3 ★ prime	4	5 ★ prime	6	7 ★ prime	8	9	10
11 ★ prime	12	13 ★ prime	14	15	16	17 ★ prime	18	19 ★ prime	20
21	22	23 ★ prime	24	25	26	27	28	29 ★ prime	30
■ PRIME (gold)		— Multiples of 2 (pink)		— Multiples of 3 (lilac)		— Multiples of 5 (green)		Special: 1 (cream)	

Complete Prime List — 1 to 30:

2	3	5	7	11	13	17	19	23	29
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10 prime numbers between 1 and 30 ✓

SECTION 6 — CLASSROOM PRACTICE (20–24 min)

Students complete the following activities independently or in pairs.

Practice Set A — Prime or Composite?

Circle P (Prime) or C (Composite) for each number. Show your reasoning.

9	P / C	Answer: C — $9 = 1 \times 9, 3 \times 3$ — 3 factors, so COMPOSITE
11	P / C	Answer: P — $11 = 1 \times 11$ only — PRIME
15	P / C	Answer: C — $15 = 1 \times 15, 3 \times 5$ — COMPOSITE
13	P / C	Answer: P — $13 = 1 \times 13$ only — PRIME
21	P / C	Answer: C — $21 = 1 \times 21, 3 \times 7$ — COMPOSITE
17	P / C	Answer: P — $17 = 1 \times 17$ only — PRIME

Practice Set B — Complete the Sieve for 1–30

Using the blank number grid (printable), students apply all 5 sieve steps step-by-step. Check: you should end up with exactly 10 primes circled.

- Step 1: Cross out 1
- Step 2: Circle 2, cross out multiples of 2 (4, 6, 8... 30)
- Step 3: Circle 3, cross out remaining multiples of 3 (9, 15, 21, 27)
- Step 4: Circle 5, cross out remaining multiples of 5 (25)
- Step 5: Circle all remaining numbers — these are ALL prime!
- Check: Did you get 2, 3, 5, 7, 11, 13, 17, 19, 23, 29?

SECTION 7 — PRIME NUMBERS IN REAL LIFE (24–27 min)

Prime numbers appear all around us! Once you know what to look for, you'll spot them everywhere.

■ Rotis	11	11 rotis on a plate — can't divide equally except 1×11
■ Holi	13	13 water balloons for Holi — can't share equally at all!
■ Cricket	7	7 cricket stumps arrangement — only 1×7 works
■ Diwali	7	Arrange 7 diyas in equal rows — impossible! 7 is prime
■ Coins	2	The ■2 coin represents the only even prime number
■ Calendar	31	31 days in a month — 31 is prime!

SECTION 8 — PRIME NUMBER PATTERNS (27–28 min)

- **After 2, ALL primes are ODD:** 2 is the ONLY even prime — every other even number is divisible by 2.
- **2 is the smallest prime:** And the most special — the only even member of the prime family!
- **Twin Primes:** Pairs of primes that are 2 apart: (3,5), (5,7), (11,13), (17,19), (29,31)...
- **Primes go on forever:** Mathematicians have found prime numbers with MILLIONS of digits — the search never ends!
- **Indian Mathematics:** Aryabhata (476 AD) and Brahmagupta (598 AD) were Indian mathematicians who also studied numbers!

SECTION 9 — KNOWLEDGE CHECK QUIZ (28–30 min)

Name: _____ Date: _____ Score: _____ / 5

Q1. Which of the following BEST describes a prime number?

A) A number divisible by many numbers

B) A number with exactly 2 factors: 1 and itself

C) A number that is always even

D) A number greater than 10

Answer: B — Prime = exactly 2 factors: 1 and itself

Q2. Which number is PRIME?

A) 9

B) 15

C) 13

D) 21

Answer: C — $13 = 1 \times 13$ only; $9 = 1, 3, 9$; $15 = 1, 3, 5, 15$; $21 = 1, 3, 7, 21$

Q3. When using the sieve, after circling 2, we should:

A) Cross out numbers ending in 2

B) Cross out all multiples of 2

C) Circle the number 4

D) Cross out the number 1

Answer: B — Cross out ALL multiples of 2 (4,6,8...30)

Q4. The number 1 is:

A) A prime number

B) A composite number

C) Neither prime nor composite

D) The smallest prime

Answer: C — 1 is special: neither prime nor composite. It stands alone!

Q5. Which of the following is the ONLY even prime number?

A) 1

B) 4

C) 6

D) 2

Answer: D — 2 is the ONLY even prime. All other primes are odd.

WHAT WE LEARNED TODAY — RECAP

Prime Number	✓ Has EXACTLY 2 factors: 1 and itself
Number 1	✓ NEITHER prime nor composite — stands alone!
Number 2	✓ The ONLY even prime number — very special!
Sieve Method	✓ 5 steps to find ALL primes in any range
Eratosthenes	✓ Ancient Greek mathematician, ~2,200 years ago
Primes 1–30	✓ 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 — 10 primes total
Memory Hook	✓ "Ek toh ek hi hota hai!" — 1 is neither prime nor composite

TEACHER NOTES & FACILITATION GUIDE

Hook Activity	Use physical laddoo-shaped objects (balls, counters) to show the sharing problem visually.
Sieve Activity	Print the 1–30 grid for each student. Use red pencil = cross out, green circle = prime.
Misconception 1	"Is 1 prime?" — Always answer: No! 1 has only ONE factor. Primes need EXACTLY 2.
Misconception 2	"Are all odd numbers prime?" — No! 9, 15, 21, 25, 27 are odd but composite.
Misconception 3	"Is 2 composite because it's even?" — No! 2 is the only even prime. Remind students often.
Cultural Hook	Ask students to think of prime numbers in their home: door numbers, page numbers, etc.
Extension (Fast Learners)	Challenge: extend the sieve to 50. How many primes are there?
Assessment	MCQ quiz can be used as a 3-minute exit card. Minimum pass: 3/5 correct.
Next Session Preview	Larger primes, prime factorisation, and why primes matter in real-world security.

COMMON MISCONCEPTIONS & CORRECTIONS

Misconception	Correct Understanding	How to Respond
"1 is prime"	1 is neither prime nor composite.	"It has only 1 factor — itself! Primes need 2 factors."
"All odd numbers are prime"	Some odd numbers are composite.	" $15 = 3 \times 3$. It's odd but has 3 factors!"
"2 is composite (even)"	2 is the ONLY even prime	" $2 \div 1 = 2$, $2 \div 2 = 1$ — only 2 ways. That makes it prime!"
"Multiples of 2 include 2"	Only 4, 6, 8... are crossed (not 2)	"CIRCLE 2 first, THEN cross its multiples."

HOMEWORK / TAKE-HOME ACTIVITY

- ♦ Apply the Sieve of Eratosthenes to find all prime numbers from 1 to 50 (printable grid provided).
- ♦ Write down 3 prime numbers you spot at home — house number, page number, number of items in a packet.
- ♦ BONUS: Ask a family member to guess if a number is prime. Teach them the 2-factor rule!
- ♦ CREATIVE: Draw the number 2 wearing a crown — "The Only Even Prime King!"
- ♦ DIWALI LINK: If you arrange 17 diyas in equal rows, what happens? (17 is prime — it refuses!)

WHAT'S NEXT — SESSION 2 PREVIEW

- Finding prime numbers up to 100 using an extended sieve.
- Prime factorisation — breaking composite numbers into their prime building blocks.
- Introduction to the concept: every number is made of prime numbers!
- More Prime Detective challenges with Aarav and Priya!

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Every child deserves to love mathematics — one prime discovery at a time. ■